



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data collection, wrangling, and formatting
 - Exploratory data analysis
 - Interactive data visualization
 - Machine learning prediction
- Summary of all results
 - Launch site and payload mass influence landing success rate.
 - Decision Tree model (94.44% accuracy) is the most effective predictor.
SVM and K-Nearest Neighbors: 83.33% accuracy.

Introduction

- Project background and context
 - Falcon 9 rocket launches by SpaceX cost 62 million dollars, but other providers cost upward of 165 million dollars; much of the savings is because SpaceX can reuse the first stage
 - If we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch
- Problems you want to find answers
 - Predict if the Falcon 9 first stage will land successfully

Section 1

Methodology

Methodology (1)

Executive Summary

- Data collection methodology:
 - SpaceX API, Web scraping from Wikipedia
- Perform data wrangling
 - Handling missing values, standardizing formats, and extracting key features.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash

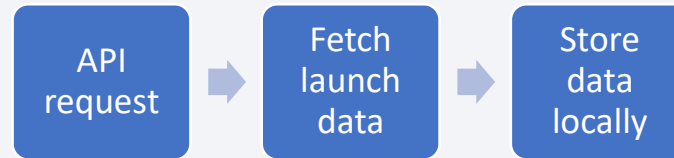
Methodology (2)

Executive Summary

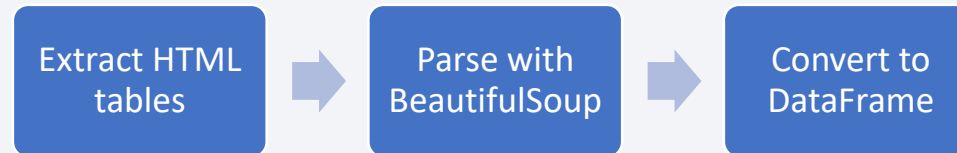
- Perform predictive analysis using classification models
 - Built various classification models: Logistic Regression, SVM, KNN, and Decision Trees.
 - Tune the best hyperparameters by GridSearchCV.
 - Evaluated classification models by the accuracy, and identified the best performing model for predicting landing success.

Data Collection

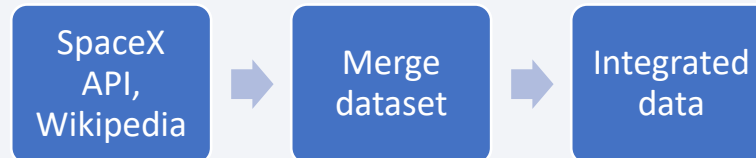
- Library: SpaceX API



- Web scraping from Wikipedia



- Data integration



Data Collection – SpaceX API

1. Initiate API Request

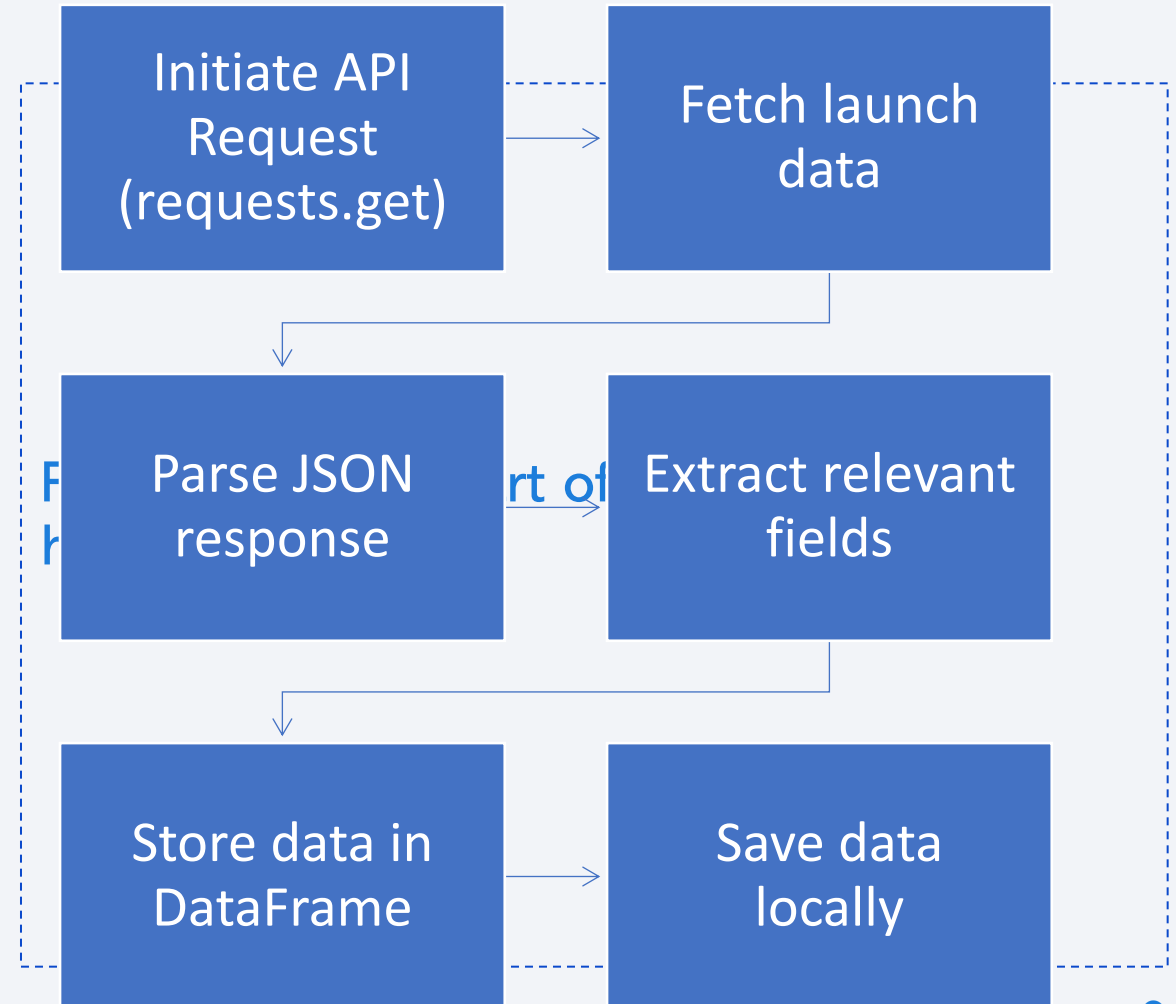
- Use Python's 'requests' library to connect to the SpaceX API.

2. Parse API Response

- Convert from JSON to a Python dictionary.
- Extract relevant fields

3. Store Data Locally

- Save extracted data into a pandas DataFrame.



Data Collection - Scraping

1. Initiate Web Scraping

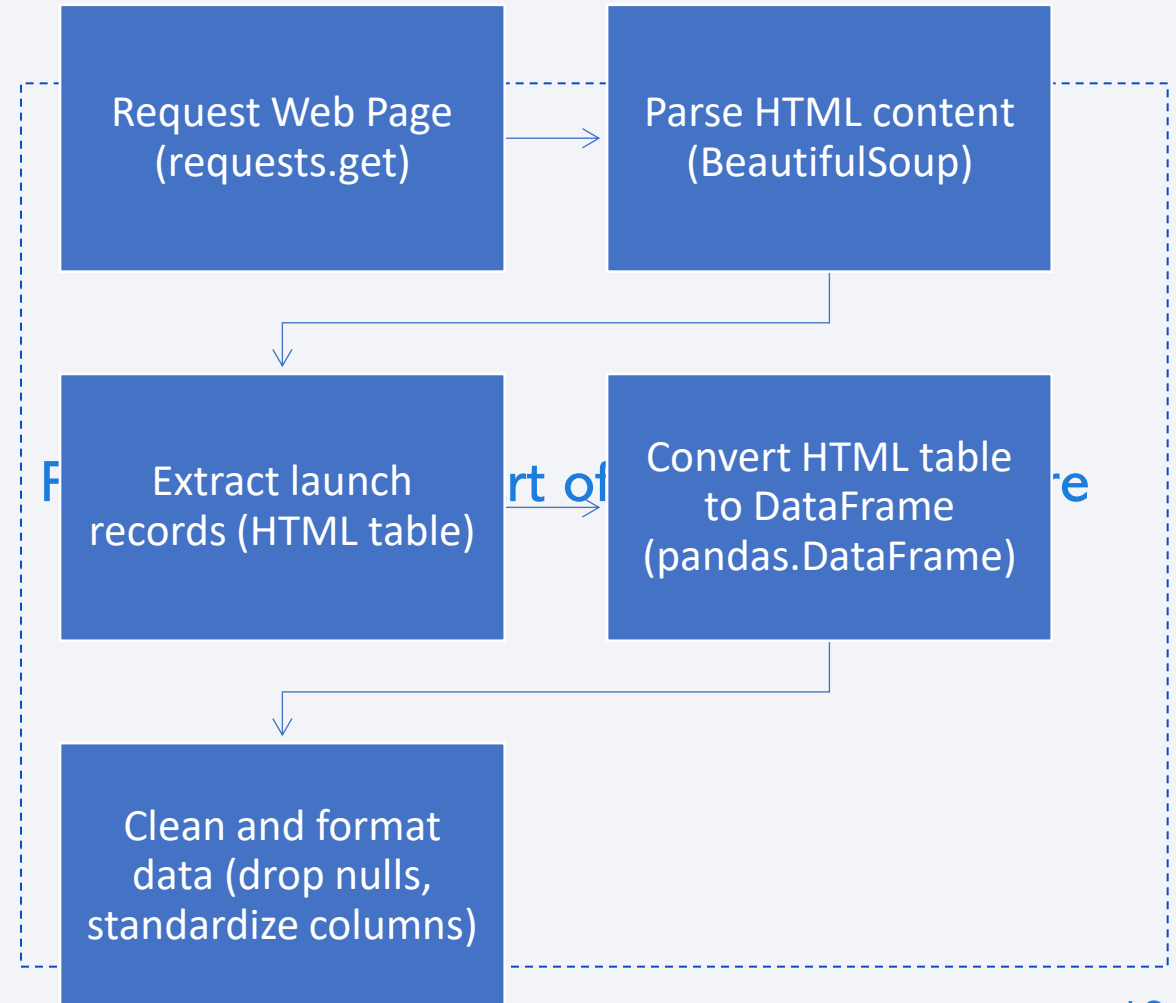
- Scraping Use Python's `requests` library to fetch the HTML content of the Wikipedia page.

2. Parse HTML Content

- Use 'BeautifulSoup' to parse the HTML content.

3. Convert to DataFrame

- Clean and format the DataFrame, ensuring data consistency.



Data Wrangling

1. Data Cleaning

- Identify and fill or remove missing values
- drop rows/columns with excessive missing data

2. Data Transformation

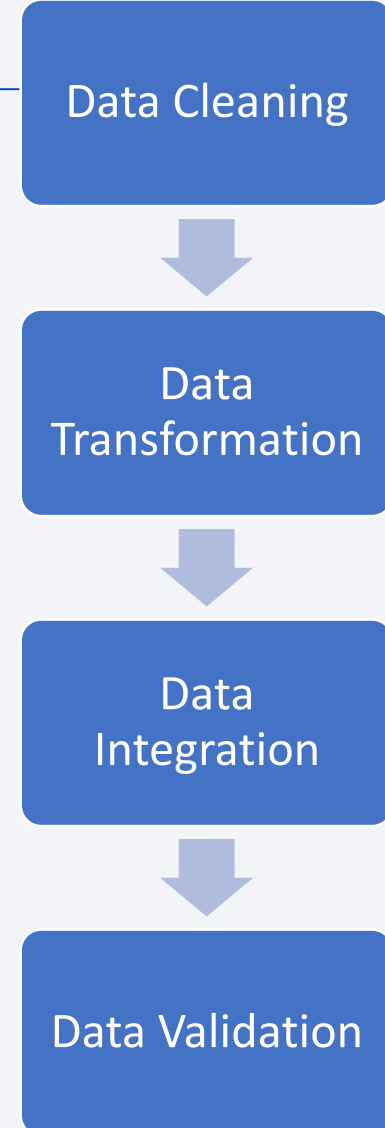
- Convert data types to appropriate formats (e.g., date-time, numerical)
- Standardize text (e.g., lowercase, remove whitespace)
- Create new features from existing data (e.g., extract year from date)
- Normalize/scale numerical features to ensure consistency

3. Data Integration

- Merge datasets collected from different sources (API, web scraping)
- Ensure consistent column names and data formats across datasets

4. Data Validation

- Remove duplicate records
- Verify the accuracy and consistency of data entries



EDA with Data Visualization (1)

1. Histograms

- visualize the distribution of numerical variables: the data spread and central tendency
- identify outliers, and assessing data skewness

2. Bar Charts

- compare categorical variables: launch outcomes (success/failure) across different categories (launch sites, rocket types)
- clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends

3. Line Charts

- track trends over time and reveal temporal patterns : the success rate of Falcon 9 launches over years
- understand performance trends or changes over specific periods

EDA with Data Visualization (2)

4. Scatter Plots

- explore relationships between two numerical variables: payload mass vs. launch success.
- identify correlations or dependencies between variables

5. Heat Maps

- visualize correlation matrices between multiple numerical variables.
- identify strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity

6. Box Plots

- display the distribution of numerical data through their quartiles
- visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories

EDA with SQL (1)

- Aggregate Queries
 - Calculated total number of launches
 - Counted successful and failed launches
 - Calculated success rates by launch site and rocket type
- Join Queries
 - Joined tables to link launch records with additional data (e.g., rocket details)
 - Combined datasets for comprehensive analysis
- Filtering Queries
 - Filtered data to focus on specific launch outcomes (success/failure)
 - Applied conditions to extract launches based on criteria: launch date or rocket configuration

EDA with SQL (2)

- Sorting Queries
 - Sorted data to identify trends or outliers
 - Ordered launches by date or success rate for analysis
- Subqueries
 - Nested queries to calculate derived metrics (e.g., average payload mass per launch site)
 - Subqueries used to perform detailed analysis within larger datasets

Build an Interactive Map with Folium

- Markers
 - Indicate launch sites: Spatial visualization of the launch locations
- Circles
 - Add circles around launch sites to visually represent proximity zones
 - Illustrates the potential impact zones, safety perimeters or operational boundaries
- Lines
 - Draw lines to connect launch sites with their proximities or other relevant locations
 - Lines provide spatial context and connections between different points of interest
 - Shows connections or relationships between launch sites and relevant features

Build a Dashboard with Plotly Dash

- Plots/Graphs

- Success Pie Chart: the distribution of successful and failed launches
→ visualize the overall success rate and performance trends
- Success-Payload Scatter Plot: payload mass & launch success relationship
→ allow users to explore how payload mass influences mission outcomes.

- Interactions

- Launch Site Dropdown: enable users to select specific launch sites for analysis
→ Facilitates filtering and focused exploration based on geographical locations.
- Range Slider for Payload: allow users to adjust payload mass ranges dynamically
→ Offers flexibility in examining launch success concerning payload mass variations.

Predictive Analysis (Classification) (1)

1. Build Models: SVM, Decision Trees, or K-Nearest Neighbors (KNN)

- Tune hyperparameters by GridSearchCV
- Tune parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN)

2. Model Evaluation

- use cross-validation techniques to ensure robustness and generalizability
- assess model performance with metrics such as accuracy, precision, recall, and F1-score

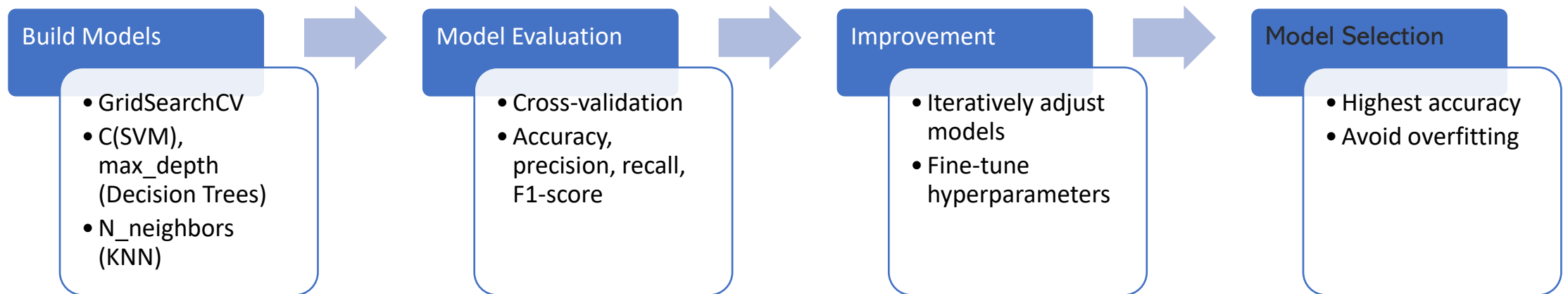
3. Improvement Iterations

- Iteratively adjust models based on insights from validation results
- Fine-tune hyperparameters to maximize predictive accuracy and reliability

4. Selection of Best Performing Model

- The highest accuracy on the test set
- Considered both training and test set performance to avoid overfitting

Predictive Analysis (Classification) (2)

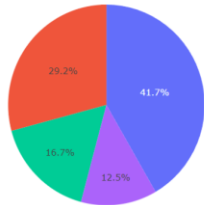


Results

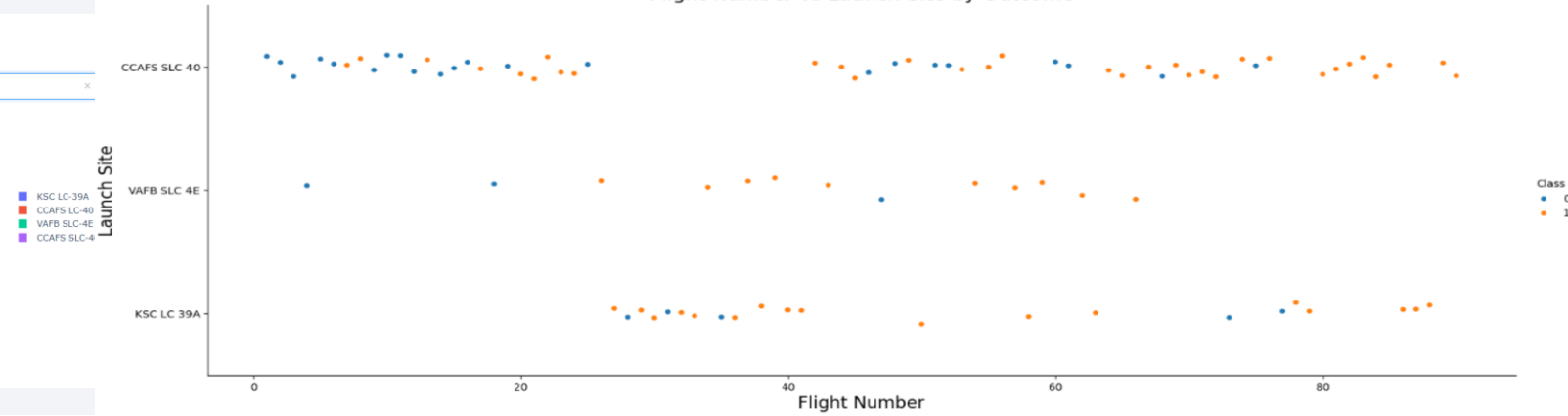
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

SpaceX Launch Records Dashboard

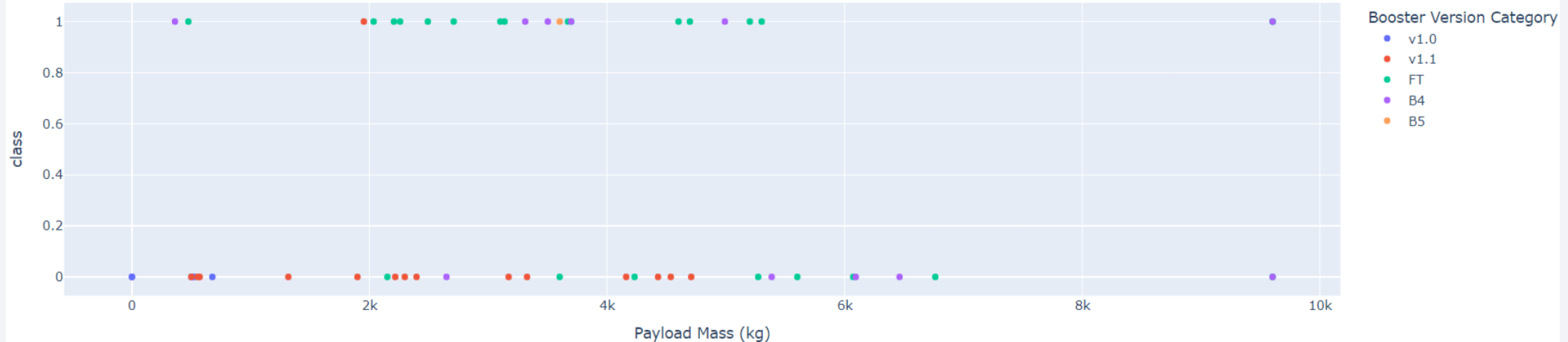
Total Success Launches by Site

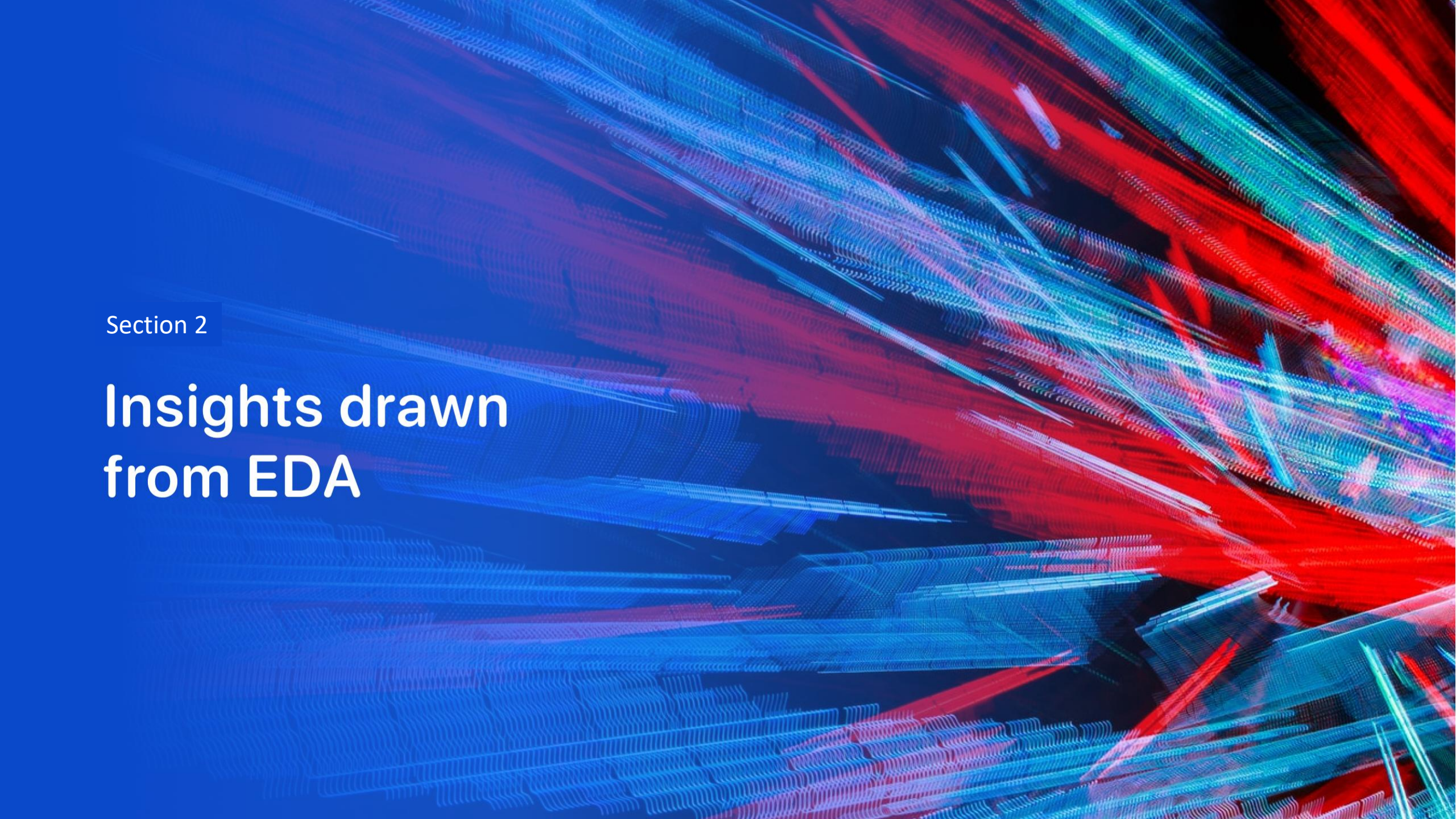


Flight Number vs Launch Site by Outcome



Payload vs. Outcome for All Sites

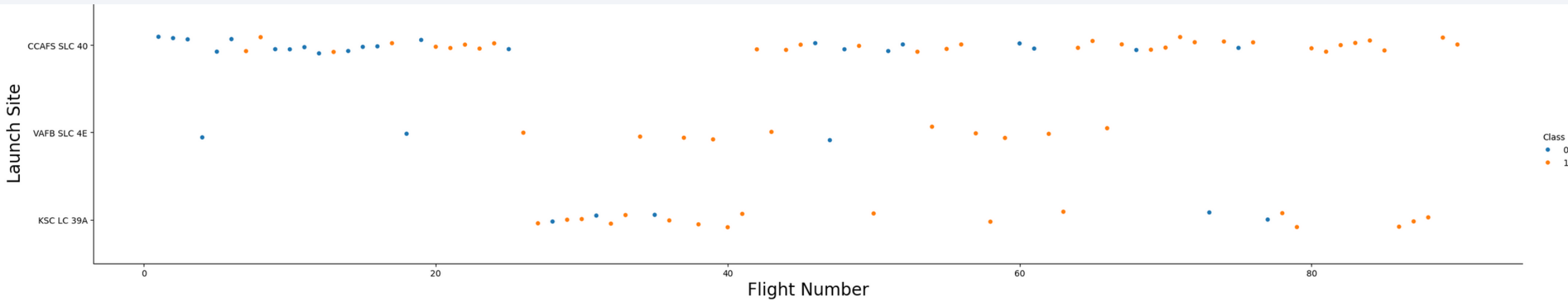


The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks vary in thickness and intensity, creating a sense of motion and depth. A faint, light blue grid pattern is visible across the entire background, adding a technical or digital feel to the design.

Section 2

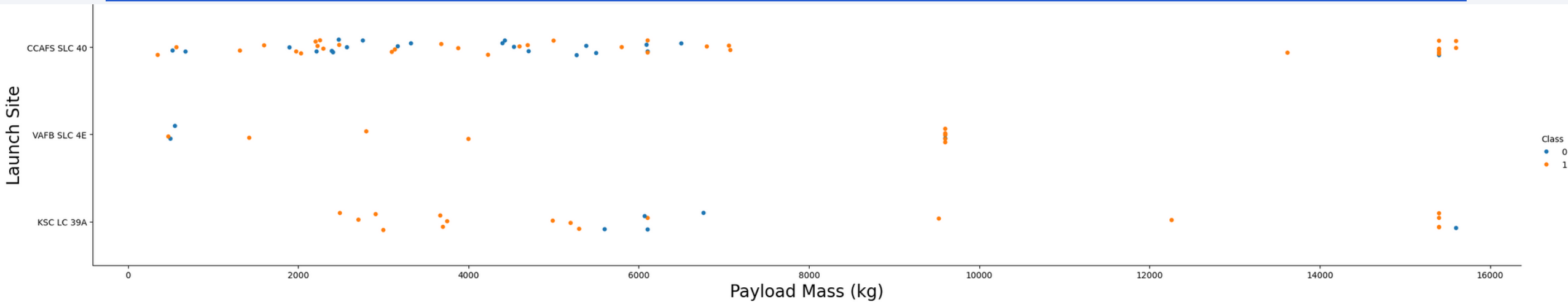
Insights drawn from EDA

Flight Number vs. Launch Site



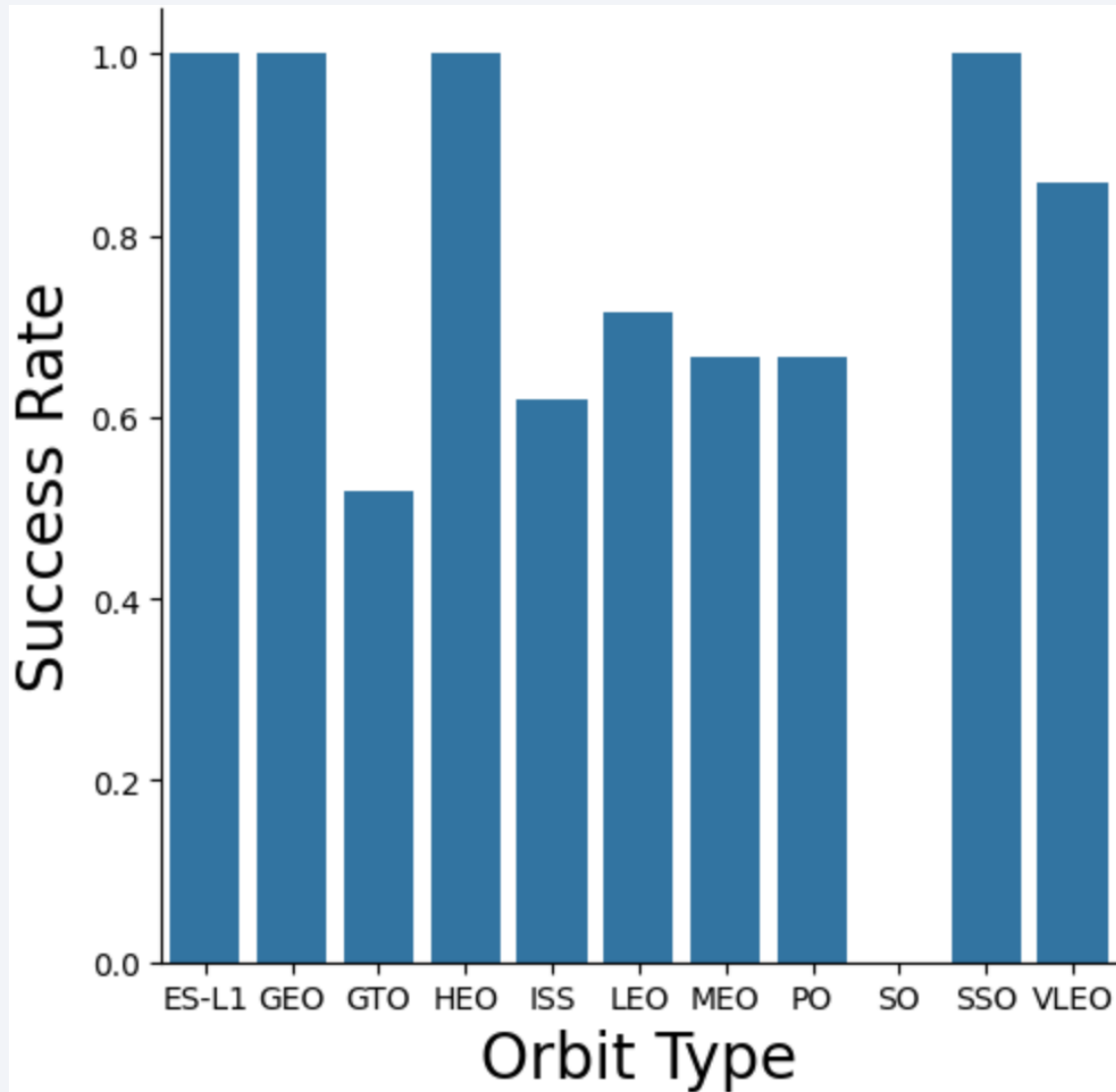
- Both CCAFS SLC 40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings
- Launches are spread across a wide range of flight numbers at all sites

Payload vs. Launch Site



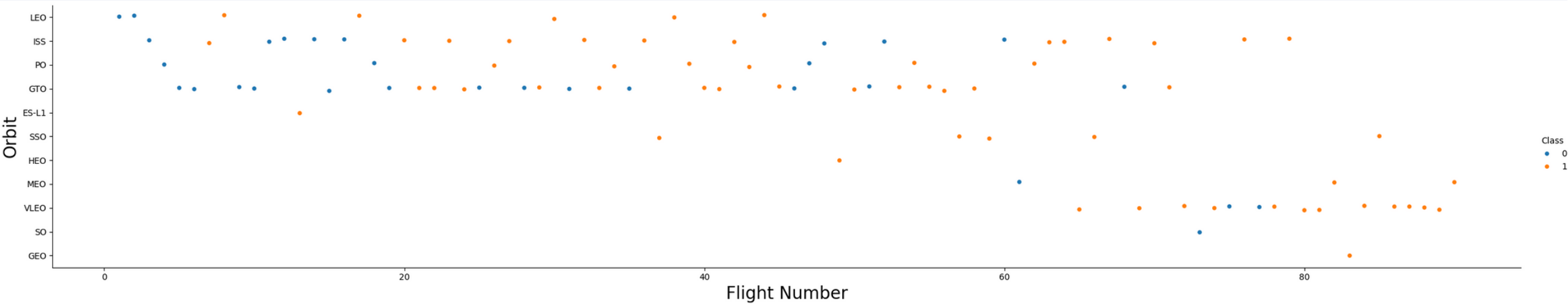
- For the VAFB-SLC launch site there are no rockets launched for heavy payload mass(greater than 10000 Kg)
- The KSC LC 39A site is frequently used for launching heavier payloads

Success Rate vs. Orbit Type



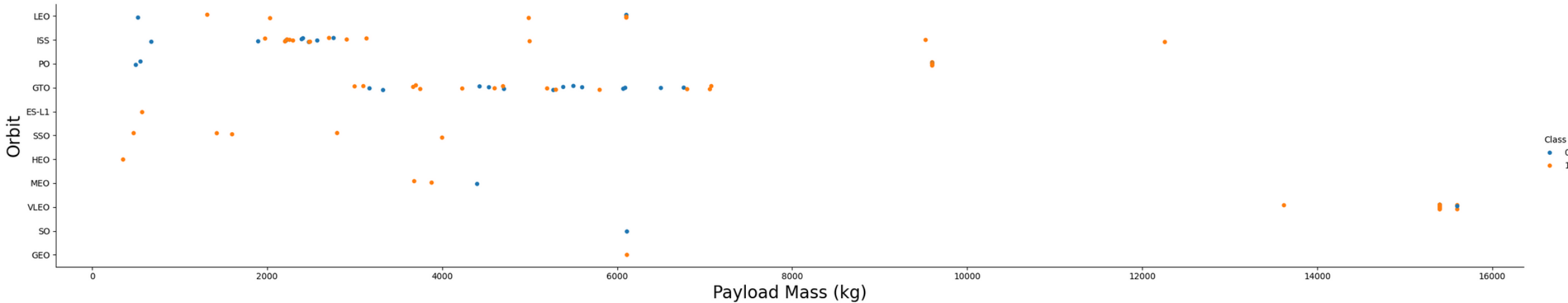
- Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have achieved a perfect success rate
- The GTO orbit type shows a significantly lower success rate compared to other orbit types

Flight Number vs. Orbit Type



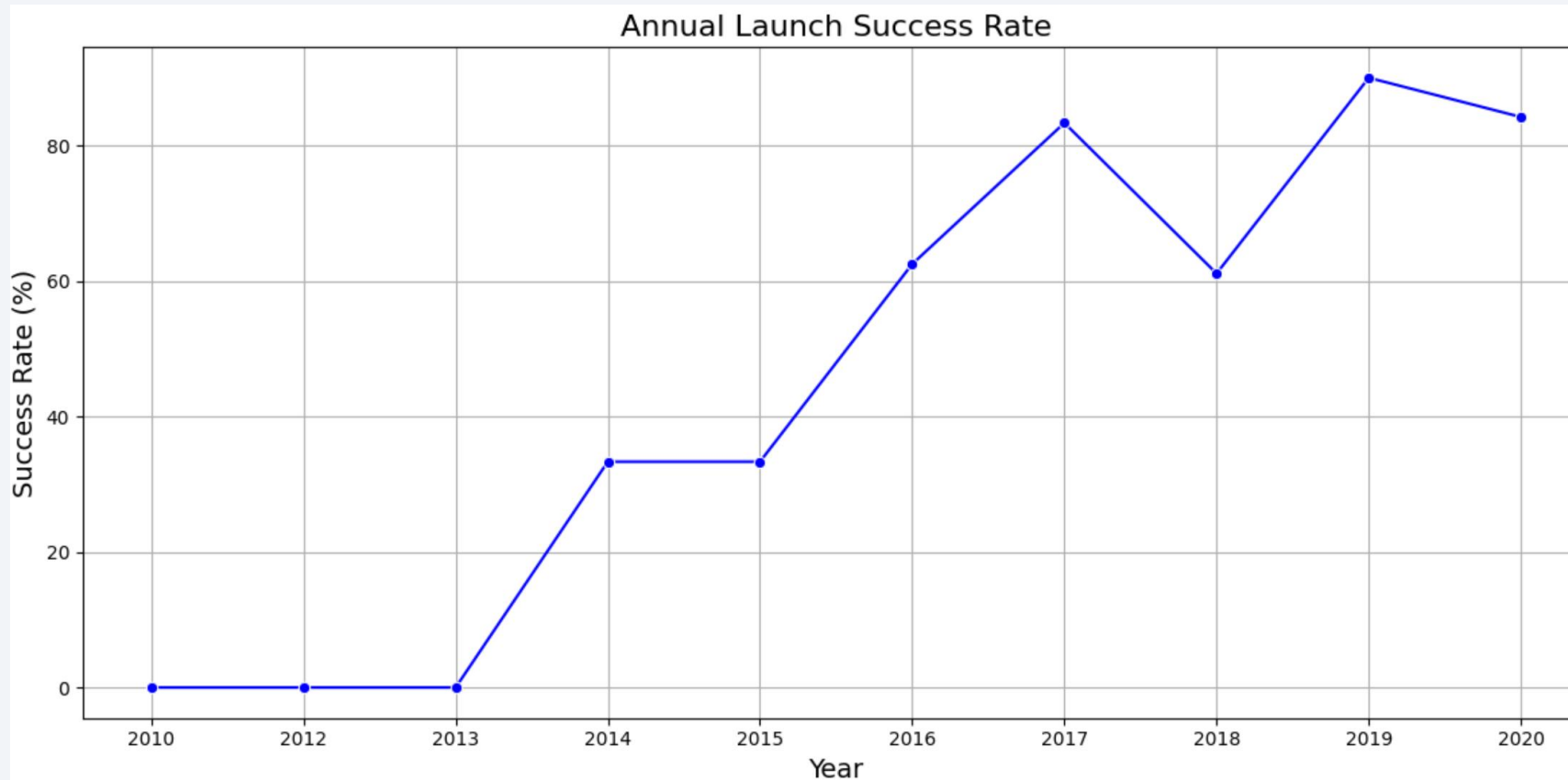
- in the LEO orbit the Success appears related to the number of flights
- there seems to be no relationship between flight number when in GTO orbit

Payload vs. Orbit Type



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS
- However for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccesful mission) are both there here

Launch Success Yearly Trend



- the success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing

All Launch Site Names

```
%sql select distinct launch_site from SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Display the names of the unique launch sites in the space mission

Launch Site Names Begin with 'CCA'

```
%sql select * from SPACEXTABLE where launch_site like 'CCA%' limit 5;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2

- Display 5 records where launch sites begin with the string 'CCA'

Total Payload Mass

```
%sql select sum(payload_mass__kg_) as total_payload_mass from SPACEXTABLE where customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

Done.

<u>total_payload_mass</u>

45596

- Display the total payload mass carried by boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

```
%sql select avg(payload_mass__kg_) as average_payload_mass from SPACEXTABLE where booster_version like '%F9 v1.1%';
```

```
* sqlite:///my\_data1.db
```

Done.

average_payload_mass

2534.6666666666665

- Display average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

```
%sql select min(date) as first_successful_landing from SPACEXTABLE where landing_outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

Done.

first_successful_landing

2015-12-22

- Use min function to Find the dates of the first successful landing outcome on ground pad

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql select booster_version from SPACEXTABLE where landing_outcome = 'Success (drone ship)' and payload_mass__kg_ between 4000 and 6000;
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

```
%sql select mission_outcome, count(*) as total_number from SPACEXTABLE group by mission_outcome;
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	total_number
-----------------	--------------

Failure (in flight)	1
---------------------	---

Success	98
---------	----

Success	1
---------	---

Success (payload status unclear)	1
----------------------------------	---

- Calculate the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

```
%sql select booster_version from SPACEXTABLE where payload_mass__kg_ = (select max(payload_mass__kg_) from SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

2015 Launch Records

```
%%sql select substr(Date, 6,2) as month, date, booster_version, launch_site, landing_outcome from SPACEXTABLE
where landing_outcome = 'Failure (drone ship)' and substr(Date,0,5)='2015';
```

```
* sqlite:///my_data1.db
```

Done.

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- SQLite does not support monthnames, so use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select landing_outcome, count(*) as count_outcomes from SPACEXTABLE
      where date between '2010-06-04' and '2017-03-20'
      group by landing_outcome
      order by count_outcomes desc;
```

```
* sqlite:///my_data1.db
```

Done.

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

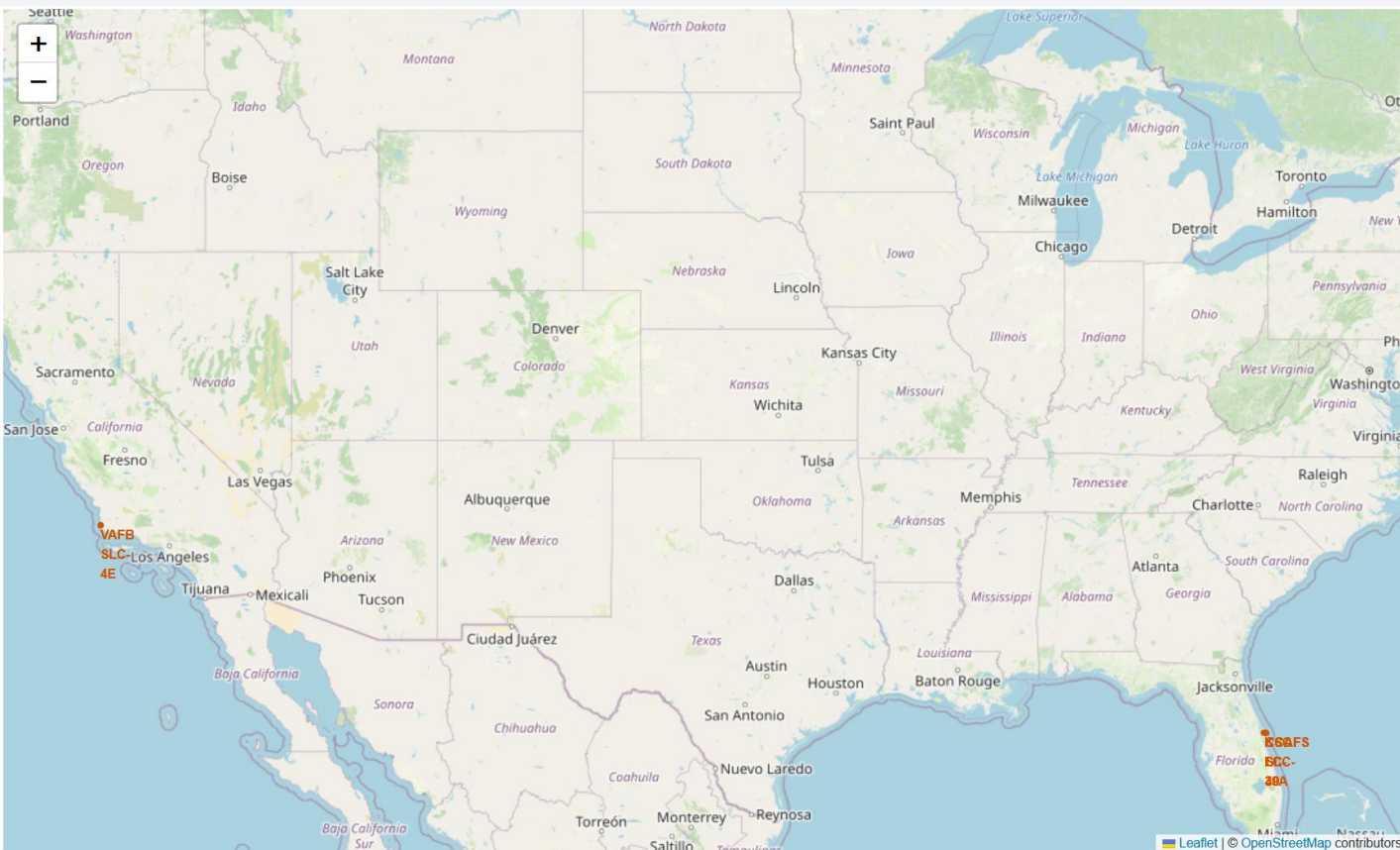
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

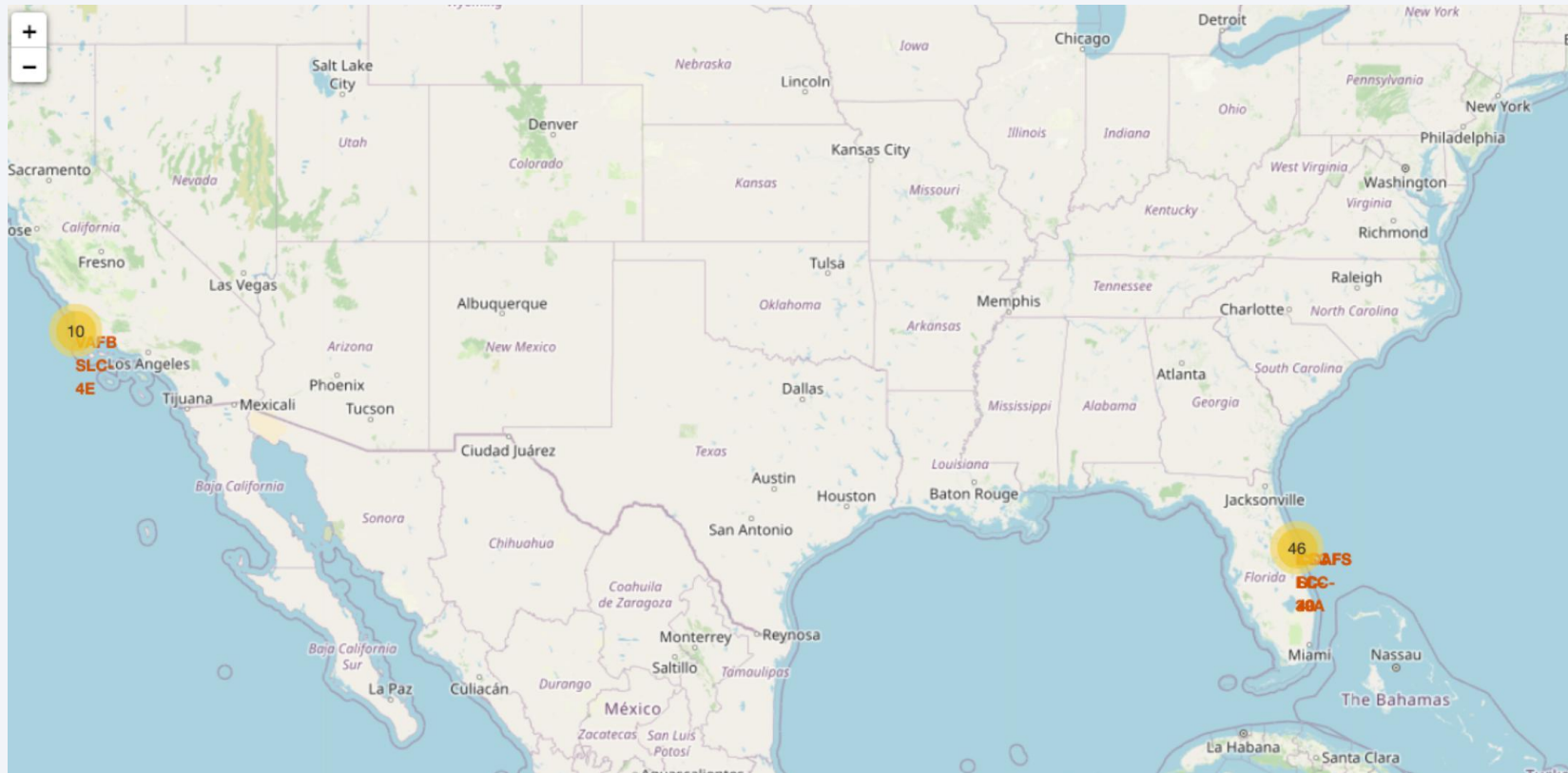
Launch Sites Proximities Analysis

Mark all Launch Sites on a Map



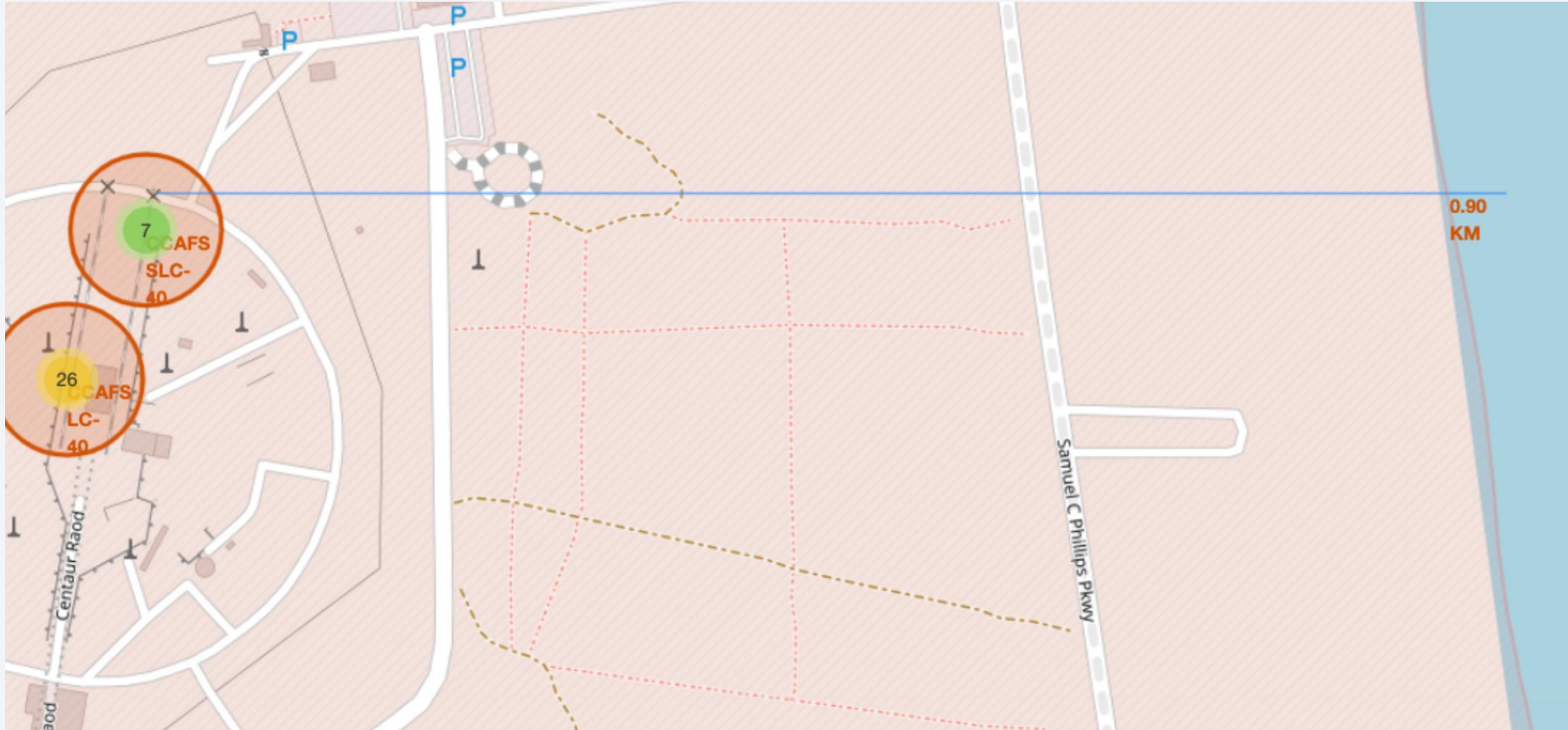
- Not all launch sites are in proximity to the Equator line
- All launch sites in very close proximity to the coast

Mark the success/failed launches for each site on the map

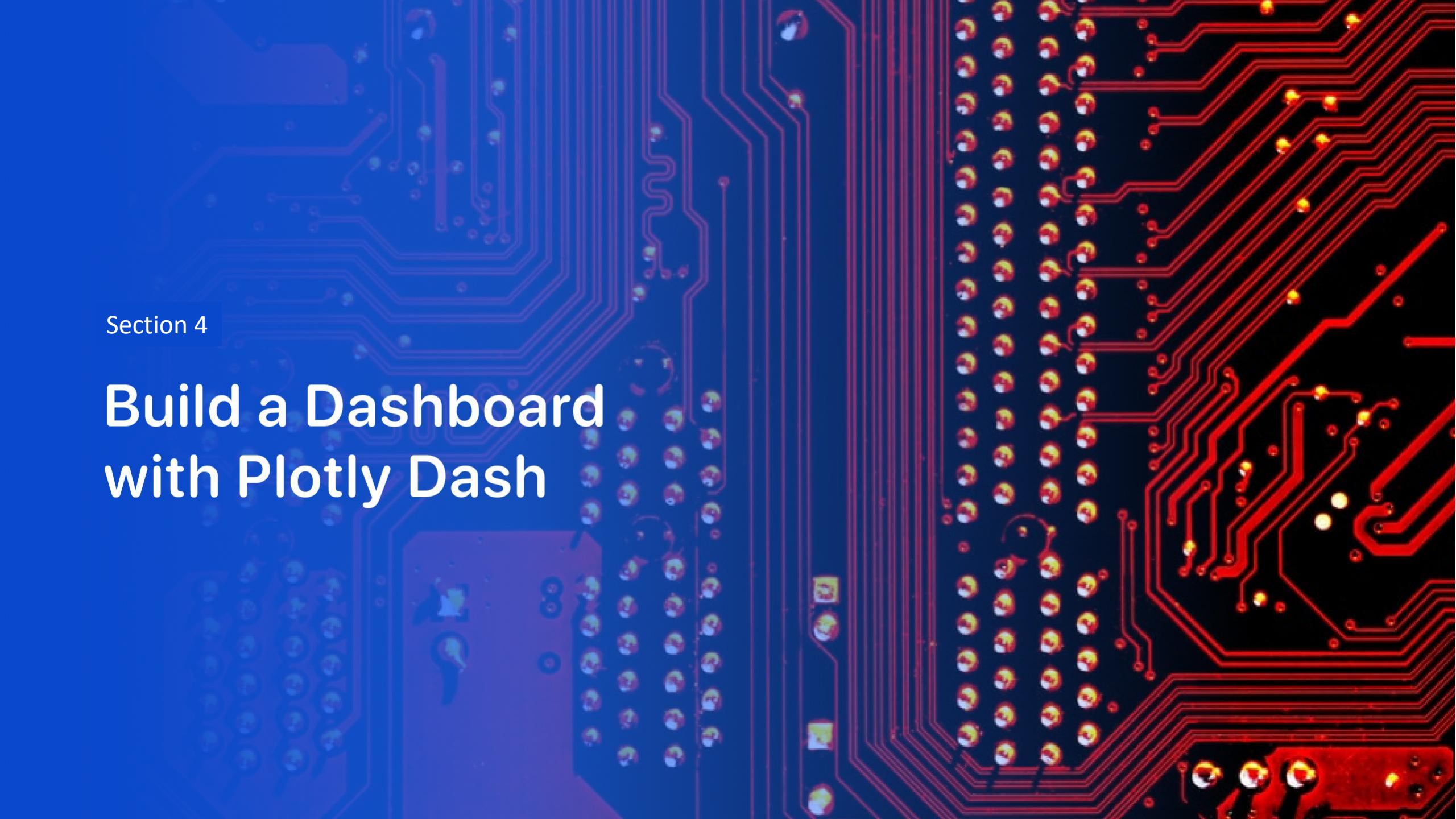


- From the color-labeled markers in marker clusters, CCAFS launch sites have relatively high success rates.

Calculate the distances between a launch site to its proximities



- the distance between the CCAFS SLC-40 launch site and the closest coastline: 0.9 km



Section 4

Build a Dashboard with Plotly Dash

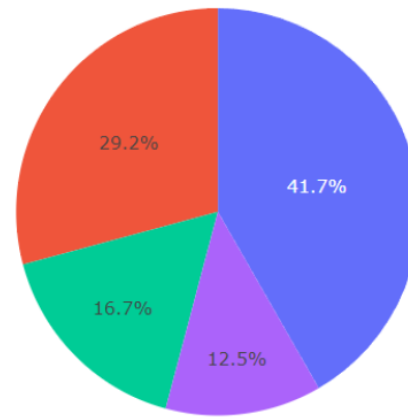
Launch Success Count for all sites

SpaceX Launch Records Dashboard

All Sites



Total Success Launches by Site



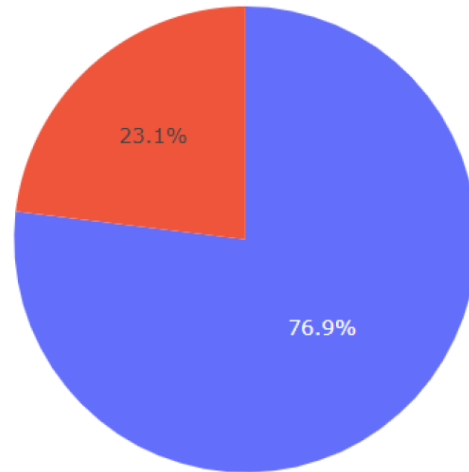
■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

Payload range (Kg):

- KSC LC-39A launch site has the highest number of successful launches (41.7%)

Launch site with highest launch success ratio

Total Success Launches for site KSC LC-39A



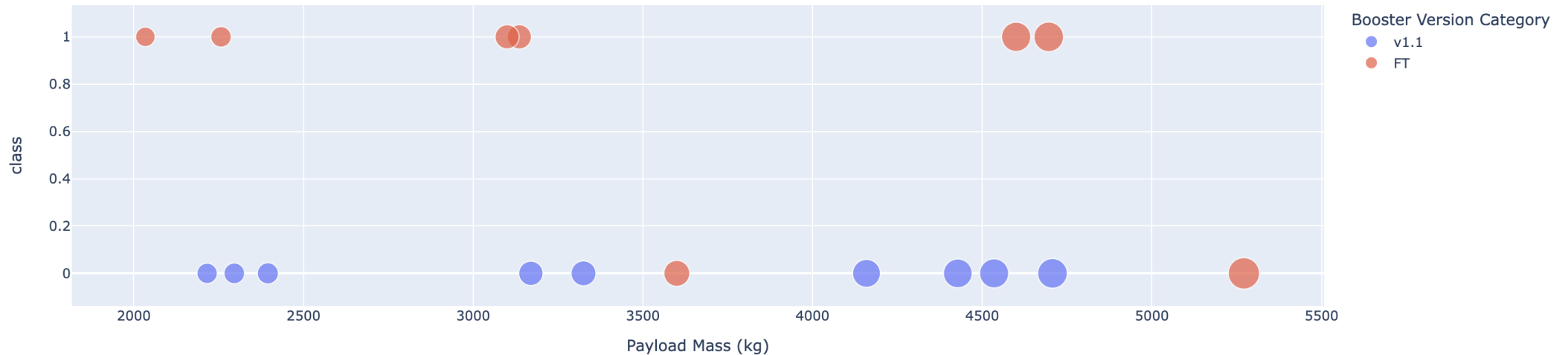
- KSC LC-39A has Class 1 (Successful Launches): 76.9%

SpaceX Launch Data Dashboard

Payload range (Kg):



Correlation Between Payload and Success for Site → CCAFS LC-40

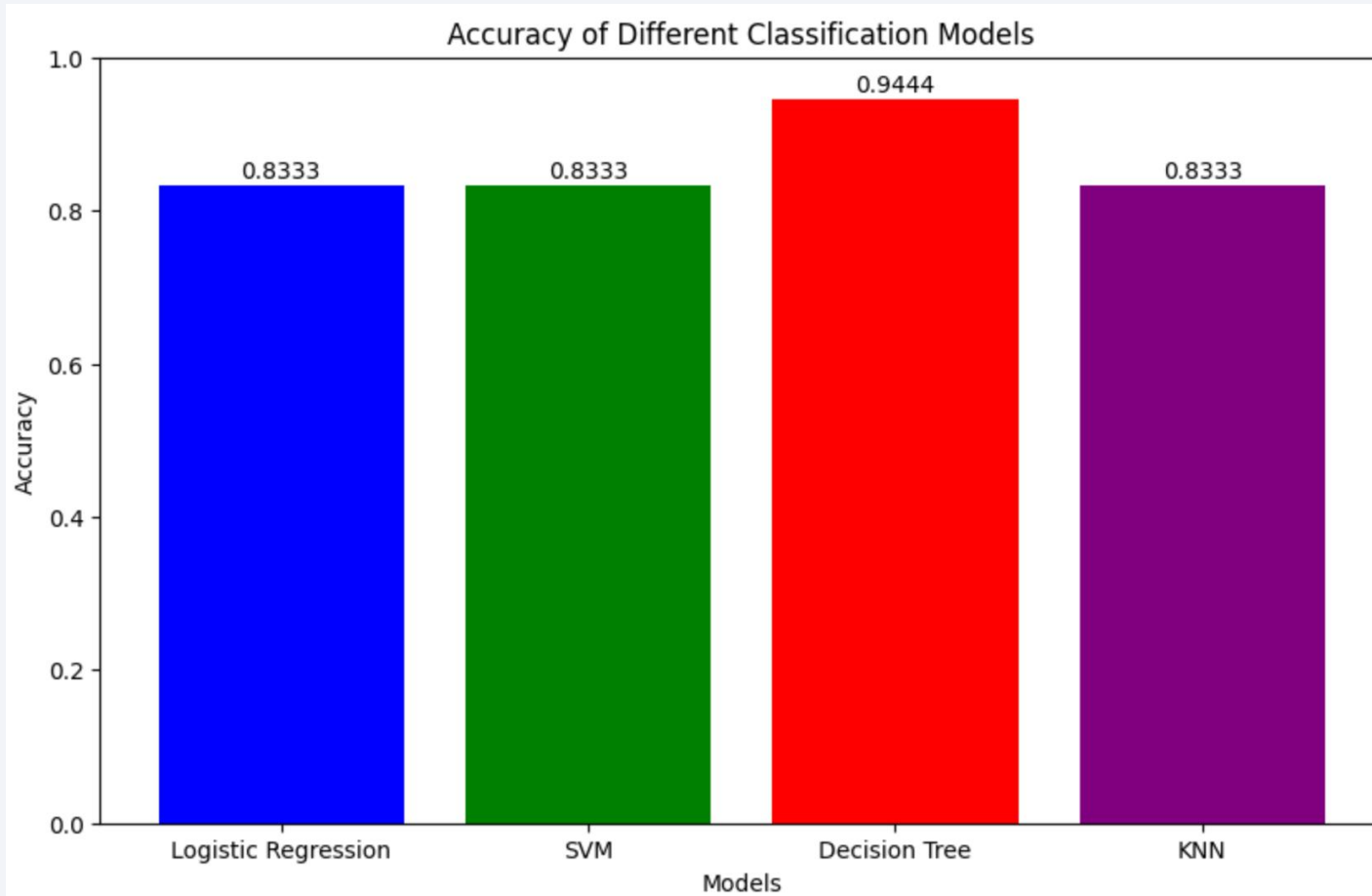


- Booster version “FT” appears to be the most frequently used and has a high success rate across various payload masses.

Section 5

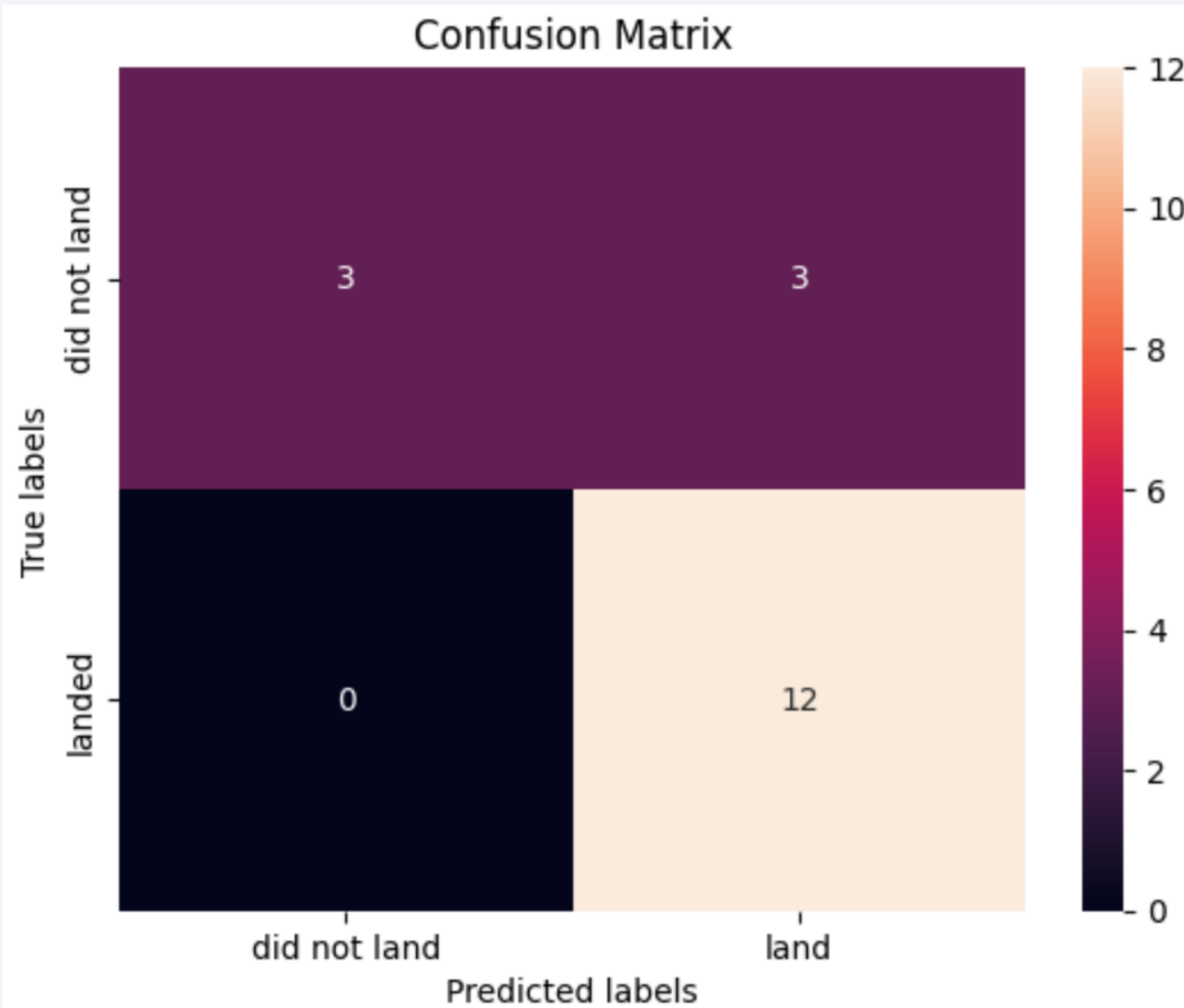
Predictive Analysis (Classification)

Classification Accuracy



- Decision Tree has the highest classification accuracy

Confusion Matrix



- The confusion matrix of the best performing model (Decision Tree): high numbers of true positives and true negatives

Conclusions

- "CCAFS LC-40" launch site has the highest success rate: 43.7%
- "FT" booster version has a high success rate across various payload masses
- No clear pattern was observed linking higher payload masses to lower success rates
- Interactive data visualizations using Folium and Plotly Dash provided valuable insights into the geographical and operational patterns of SpaceX launches

Thank you!

